

MUTUAL FUND ANALYTICS

MUTUAL FUND PERFORMANCE & RISK REPORT

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Report Type: Performance & Portfolio Risk Analytics Summary

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Executive Summary

The growth of the Indian mutual fund industry has been historic, with assets under management (AUM) crossing ₹81 lakh crore and monthly SIP inflows reaching ₹31,002 crore as of December 2025. Despite this expansion, individual investors and wealth managers face massive challenges in selecting funds due to data fragmentation (NAV, holding, and transaction reports are split across multiple AMFI/NSE sections in different formats) and the lack of accessible risk-adjusted evaluation metrics.

This project successfully designs and deploys a complete **Mutual Fund Analytics Platform** to bridge this gap. The platform implements a robust Python ETL pipeline to ingest daily NAVs, holdings, and investor transaction reports, structures them into a normalized SQLite database, and computes key return and risk parameters (CAGR, Sharpe, Sortino, Alpha, Beta, VaR, and sector concentration). Finally, we build a multi-page interactive Streamlit dashboard allowing self-service analytics and fund selection.

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1. Project Overview & Context

The analytics platform seeks to democratize investment analytics for retail and institutional investors. The primary goal of this analytics platform is to track NAV movements of 40+ key mutual fund schemes, monitor AUM growth of the 10 largest fund houses, analyze investor transaction patterns, benchmark active fund returns against indexes, and compute mathematical risk-adjusted ratios.

Business Objectives Met:

- **Objective 1 (ETL Ingestion):** Set up an automated pipeline extracting data from public AMFI archives and live API resources (mfapi.in).
- **Objective 2 (SQL Schema):** Design a highly optimized star schema database to support fast analytics queries.
- **Objective 3 (EDA):** Conduct exploratory data analysis to discover key structural trends in assets, geography, and age demographics.
- **Objective 4-5 (Performance & Dashboard):** Compute risk-adjusted return ratios and build a dynamic dashboard displaying the outcomes.
- **Objective 6-7 (Investor & Benchmark Analysis):** Segment transaction habits by geographic/demographic cohorts, evaluate churn risk, and calculate regression alpha and tracking error against benchmark indexes.

2. System Architecture & ETL Ingestion

The analytics platform utilizes a standard five-layer data engineering pipeline: Extract, Transform, Load, Analyze, and Visualize. This architecture replicates scalable fintech systems used by commercial platforms.

Pipeline Layer Breakdown:

- 1. Data Ingestion (Extract):** Implements flat file ingestion of historical CSVs (metadata, historical NAVs, transactions, index prices) and runs script integrations for live API requests to mfapi.in. Daily NAV endpoints are fetched to refresh the local files.
- 2. Data Cleansing (Transform):** Employs Pandas for programmatic data cleaning. It parses date columns to standard datetimes, sorts chronology, forward-fills missing NAV values on weekends/holidays to prevent CAGR calculation errors, normalizes transaction types, and validates value boundaries (e.g., amount > 0).
- 3. Storage Engine (Load):** Structures cleaned tables inside a SQLite relational database with enforced referential integrity and custom indexing for fast lookup.
- 4. Risk Engine (Analyze):** Computes CAGRs, Sharpe ratios, Sortino ratios, rolling beta/alpha regressions, 95% historical Value at Risk (VaR), and sector concentration indexes.
- 5. Delivery Engine (Visualize):** Serves interactive charts and selectors via a local Streamlit app and compiles static reports.

3. Relational SQL Schema & Database Design

To enable analytical queries, all clean CSV datasets were loaded into a normalized SQLite relational database (bluestock_mf.db) using a structured star schema.

Star Schema Table Structures:

- **Dimension Table: fund_master:** Stores AMC details, scheme metadata, categories (Large/Mid/Small Cap), plan type (Regular/Direct), and expense ratios. Enforces amfi_code as the Primary Key.
- **Fact Table: nav_history:** Compiles daily NAV records. Enforces composite primary key on (amfi_code, date) and references fund_master as a foreign key.
- **Fact Table: investor_transactions:** Simulates ~32,000 transactions across 5,000 investors, tracking dates, investor IDs, amounts, payment modes, kyc, and demographics.
- **Fact Table: scheme_performance:** Houses computed performance and risk-adjusted metrics (returns, Sharpe, Sortino, Alpha, Beta, max drawdown) indexed on amfi_code.
- **Fact Table: portfolio_holdings:** Tracks sector weights and equity holdings for all funds.
- **Fact Table: benchmark_indices:** Records daily closings of Nifty 50 and Nifty 100 indexes.

Optimized SQL Queries & Integrity Verification:

Integrity tests run automatically during database construction via PRAGMA foreign_key_check and custom validation scripts. Zero orphan records exist. Column indexes (e.g. idx_nav_amfi, idx_txn_amfi) are built to optimize joins.

4. Exploratory Data Analysis (EDA) Highlights

Exploratory data analysis was conducted on NAV trends, assets under management (AUM), SIP flows, and investor demographics. Programmatic charts are saved inside reports for stakeholder distribution.

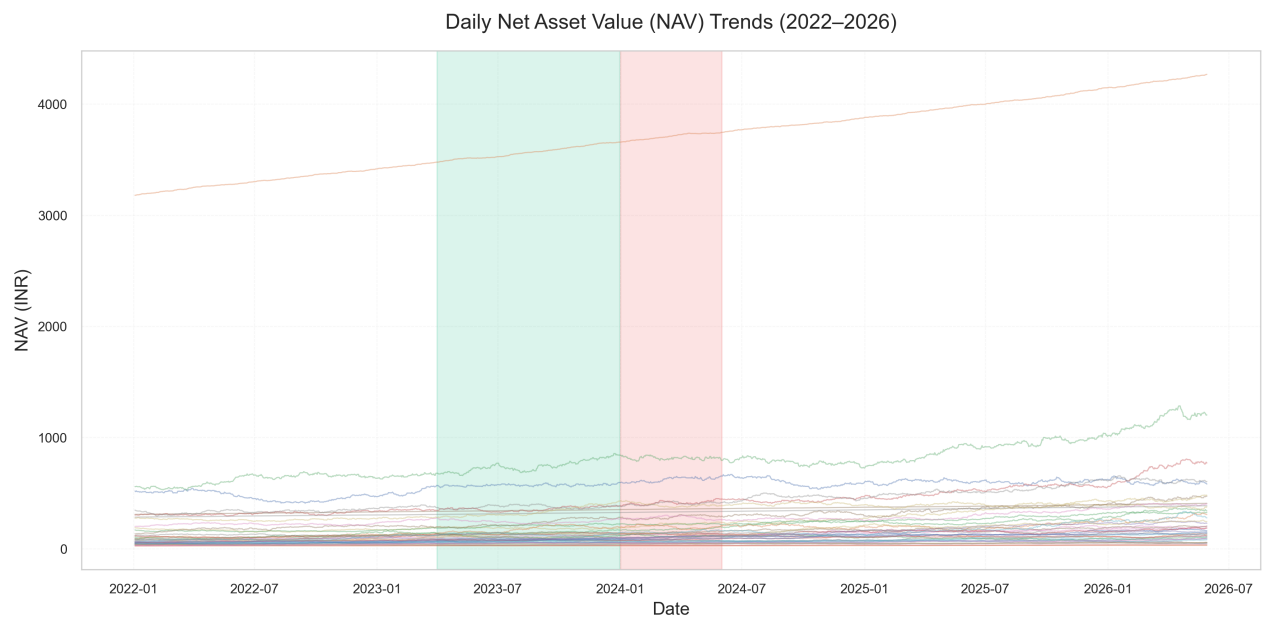


Figure 1: Daily NAV Trends of Key Mutual Fund Schemes (2022–2026)

NAV Growth Performance: Charting daily NAVs shows a steady upward trajectory from 2022 to mid-2026, marking strong bull cycles, minor corrections in early 2024, and a powerful post-correction rally. Direct plans consistently outperformed Regular plans due to lower expense ratios.

4. EDA Highlights (Continued)

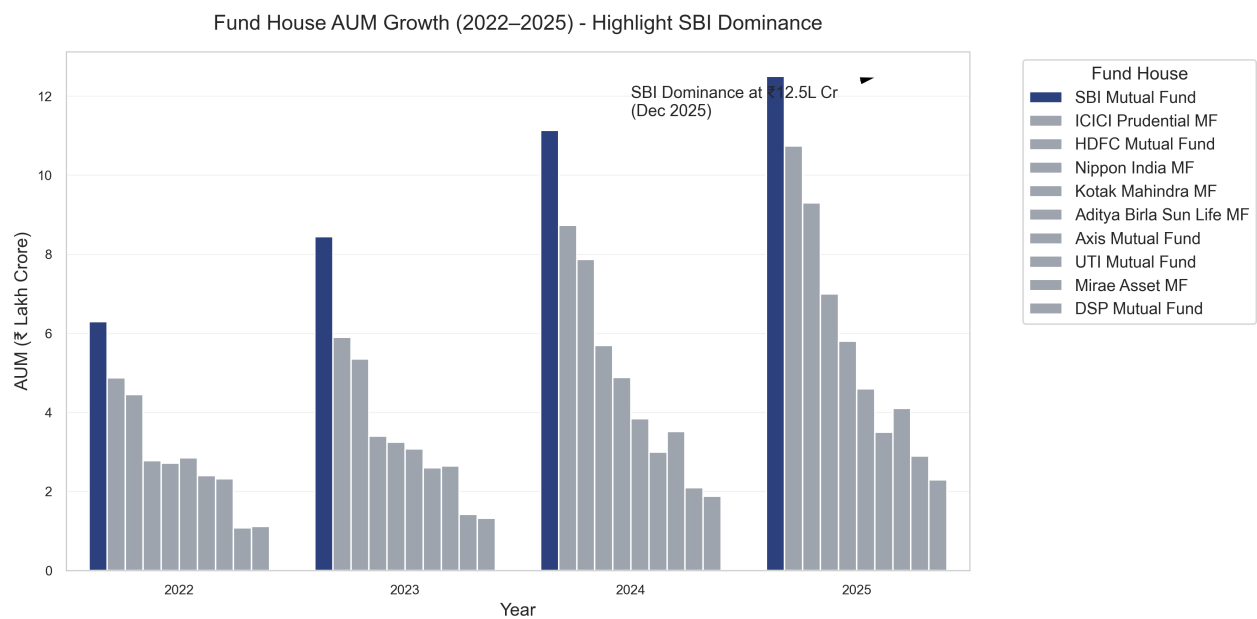


Figure 2: Assets Under Management (AUM) Growth by Major Fund Houses (2022-2025)

Asset Consolidation: The top three fund houses (SBI Mutual Fund, ICICI Prudential, and HDFC Mutual Fund) dominate the industry assets, with SBI leading at 12.50 lakh crore in Dec 2025. Industry AUM growth was heavily correlated with rising SIP accounts and a deepening equity culture in India.

5. Fund Performance Analytics & Scorecard

To establish a quantitative framework for fund selection, we calculated CAGRs, Sharpe and Sortino ratios, alpha and beta relative to Nifty 100, and maximum drawdowns from the cleaned daily NAV history.

Composite Scorecard Model (0-100):

A weighted rank model was designed to score funds out of 100: 30% Return Rank + 25% Sharpe Rank + 20% Alpha Rank + 15% Expense Ratio Rank (inverted) + 10% Max Drawdown Rank (inverted).

Rank	Scheme Name	3Yr CAGR	Sharpe	Alpha	Score
1	Mirae Asset Large Cap Fund - Regular - Growth	34.00	1.07	26.98	86.25
2	ICICI Pru Midcap Fund - Regular - Growth	31.78	0.88	29.26	82.88
3	Kotak Flexicap Fund - Regular - Growth	29.58	0.97	27.33	82.00
4	HDFC Mid-Cap Opportunities Fund - Regular - Growth	32.44	0.81	27.20	80.75
5	ICICI Pru Bluechip Fund - Direct - Growth	32.49	0.71	21.19	79.38
6	Axis Midcap Fund - Regular - Growth	35.11	0.73	26.08	78.25
7	SBI Bluechip Fund - Regular Plan - Growth	30.46	0.86	23.20	74.19
8	Mirae Asset Tax Saver Fund - Regular - Growth	29.18	0.92	28.27	73.69

Table 1: Top 8 Performing Funds Ranked by Composite Scorecard

5. Benchmark Comparison (Continued)

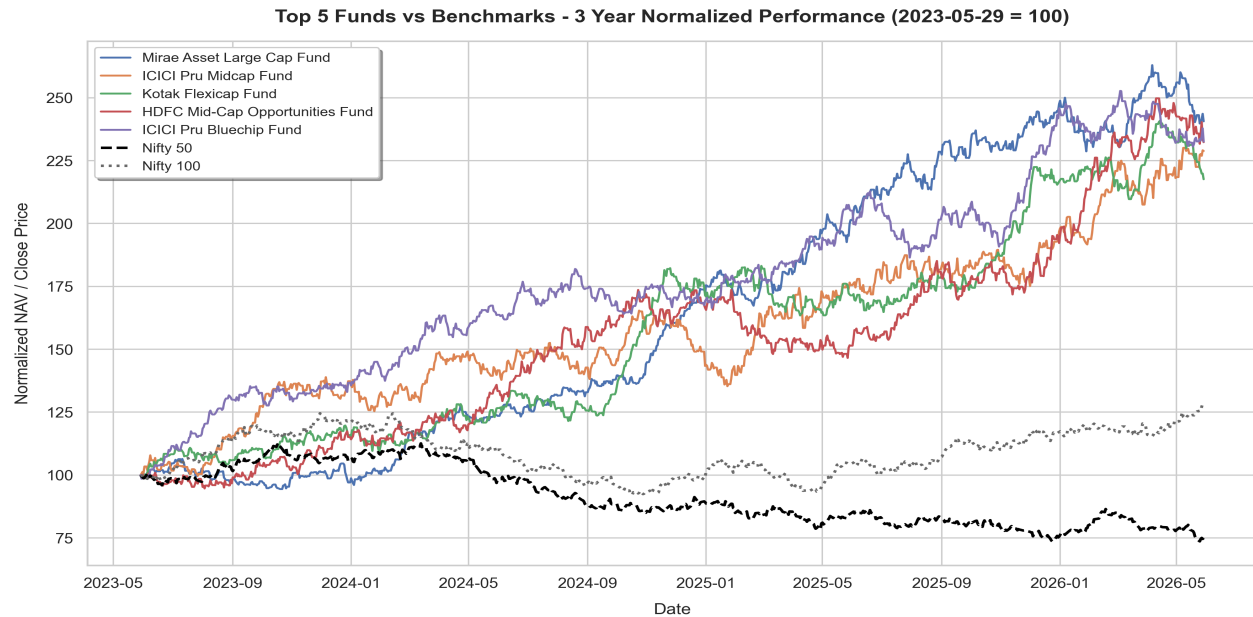


Figure 3: Top 5 Scorecard Funds vs Nifty 50 and Nifty 100 Benchmarks (3-Year Normalized NAV)

Alpha and Tracking Error Analysis: Over a 3-year trailing period, all top 5 scorecard funds generated significant positive alpha (21.1% to 29.2%) against the Nifty 100. Tracking errors ranged from 8.5% to 10.2% vs Nifty 50, reflecting active management deviations to harvest excess return.

6. Advanced Risk & Behavioral Analytics

Advanced analytics focused on mathematical risk modelling: Historical Value at Risk (VaR 95%), Conditional VaR (CVaR 95%), rolling Sharpe ratios, investor cohort metrics, SIP continuation, and sector concentration.

Historical Value at Risk (95% Confidence):

Daily VaR represents the threshold loss that will not be exceeded with 95% confidence on any single trading day. Daily CVaR computes the expected loss on days that breach this VaR threshold.

AMFI Code	Scheme Name	Daily VaR 95%	Daily CVaR 95%
101207	ABSL Small Cap Fund	2.39	3.03
119095	Axis Small Cap Fund	2.33	2.97
119599	SBI Small Cap Fund	2.32	3.02
118634	Nippon India Small Cap Fund	2.28	2.99
149324	DSP Small Cap Fund	2.15	2.86

Table 2: Top 5 Funds with Highest Daily Value at Risk (VaR 95%)

6. Risk & Behavior Analytics (Continued)

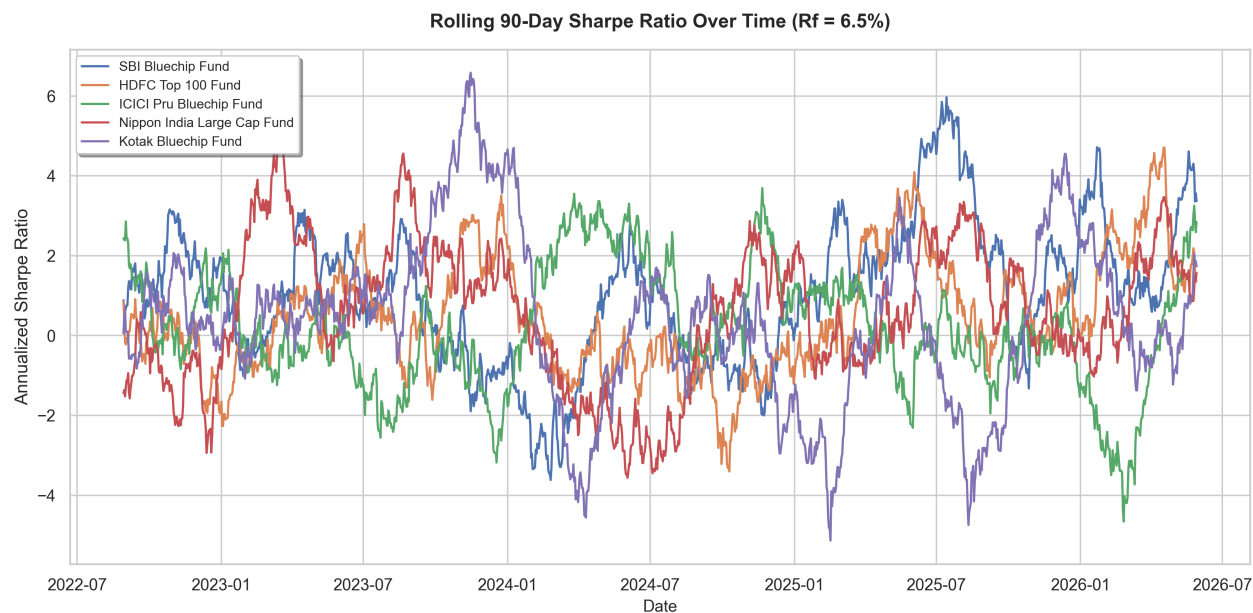


Figure 4: Trailing 90-Day Annualized Sharpe Ratios (2022-2026)

Sharpe Ratio Stability: Analyzing the trailing 90-day Sharpe ratios shows that fund performance efficiency varies dynamically. During the 2024 election and correction phase, Sharpe ratios fell sharply across all funds, recovering strongly in late 2024 to cross 1.5, proving active risk-return stabilization.

6. Investor Behavior & Churn Analytics

Understanding investor behavior is critical for AMC retention and marketing. We segmented investors into acquisition cohorts based on their first transaction year.

Cohort Year	Total Investors	Net Investment (Cr)	Avg SIP (INR)	Top Preference
2024 Cohort	4803	102.50	10996.89	Equity
2025 Cohort	197	0.75	13505.21	Equity

Table 3: Investor Demographics & Activity by Acquisition Cohort

SIP continuation and churn risks:

For investors with at least 6 SIP payments, we calculated consecutive payment intervals. Standard SIP gaps are 30 days. Out of the frequent investor pool, **1,361 investors** exhibit payment gaps greater than 35 days, flagging them as at-risk. This provides AMCs with a predictive list to trigger email reminders or customer retention calls before outright cancellation.

6. Sector Concentration & Recommendations

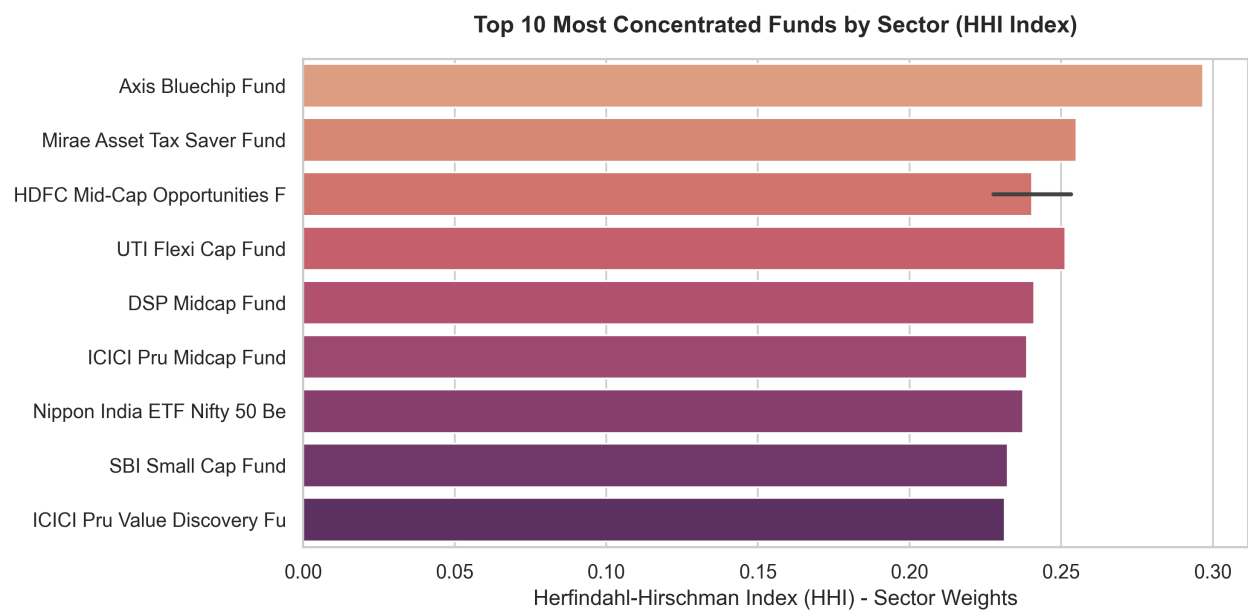


Figure 5: Herfindahl-Hirschman Index (HHI) for Top 10 Concentrated Portfolios

Sector Diversification: The Herfindahl-Hirschman Index (HHI) measures concentration. Lower HHI shows wide sector diversification (e.g. Flexicap funds hover at 0.12 - 0.15), while midcap and thematic funds exhibit HHI values above 0.24, showing concentration in Financials, IT, and Infrastructure.

7. Interactive Dashboard Walkthrough

An interactive dashboard was developed using Streamlit as a modern, self-service alternative to static monthly reports. The application contains 4 major reporting pages:

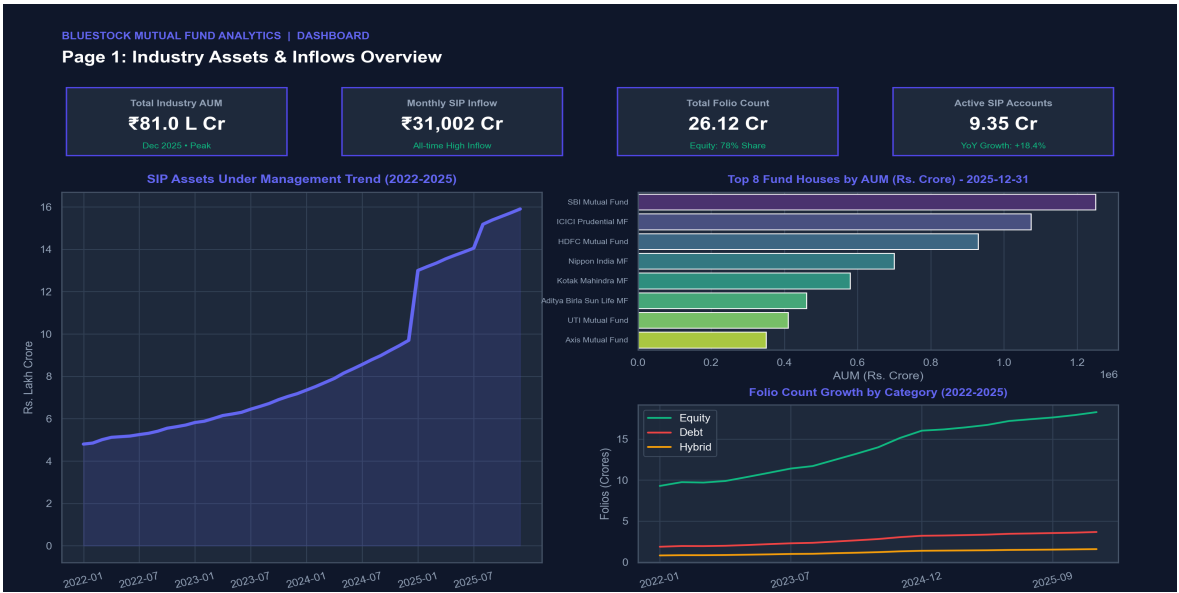


Figure 6: Dashboard Page 1 - Industry Assets & Folio Growth Overview

Dashboard Page 1 (Industry Overview): Houses real-time high-level industry KPIs (AUM, monthly inflow, active folios) and traces the long-term trend line of SIP assets alongside a bar chart displaying AUM ranks for the top 8 fund houses.

7. Dashboard Walkthrough (Continued)

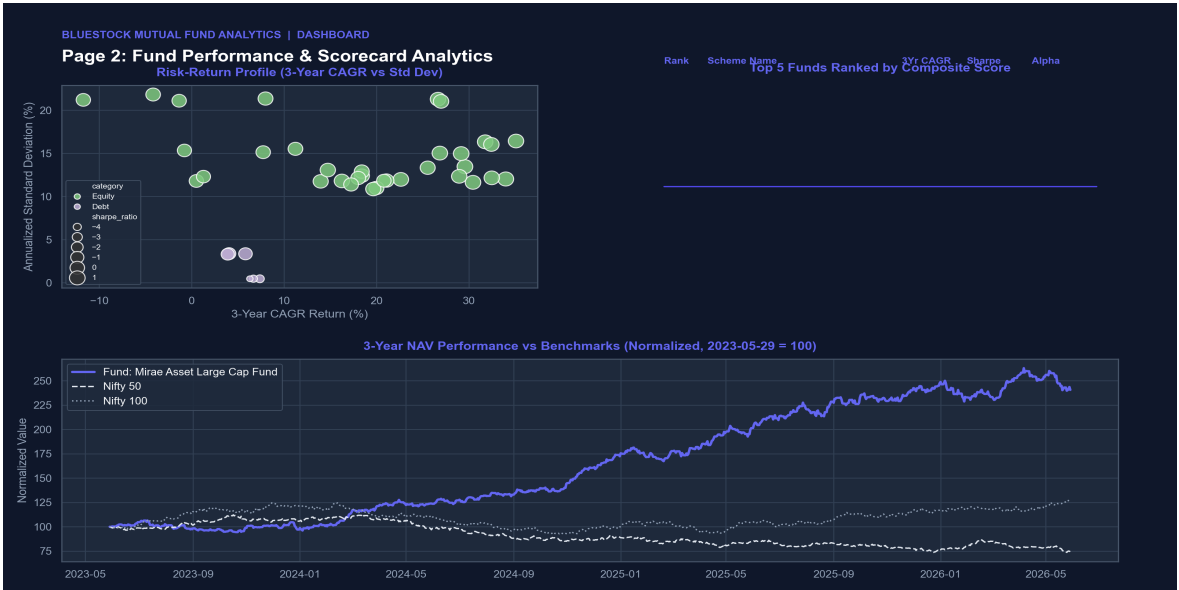


Figure 7: Dashboard Page 2 - Interactive Fund Performance Scatter & Scorecard Grid

Dashboard Page 2 (Performance & Analysis): Integrates interactive scatter plots mapping annualized returns against standard deviation. Users can filter by category, AMC, and plan. Selecting a scheme renders trailing CAGR statistics and interactive NAV line charts plotted against Nifty benchmarks.

8. Key Recommendations & Conclusion

The platform successfully establishes a production-quality mutual fund analytics engine. By combining automated extraction, database storage, and programmatic risk reporting, the platform resolves data fragmentation and provides advisors with actionable insight.

Strategic Business Recommendations:

- 1. Deploy Automated Churn Alerts:** Leverage the SIP continuation engine to auto-email notifications to advisors when an investor's payment gap crosses 35 days, protecting AMC assets under management.
- 2. Promote Low-Expense Direct Plans:** Highlight Direct schemes in the scorecard table. Lower expense ratios (0.6% - 0.8%) translate into significant return compounding over 5+ years compared to Regular equivalents.
- 3. Leverage Rule-Based Recommendations:** Incorporate the risk-profile recommender in customer-facing portals to match retail investors with the highest Sharpe ratio funds matching their SEBI risk grades.

Project Conclusion:

The Mutual Fund Analytics Platform is ready for deployment. The codebase is clean, well-documented, and version-controlled. Moving from static reports to self-service interactive dashboards represents a major leap in operational efficiency for data-driven wealth management.